1st Midterm · Unit 4 · Linear systems and phase portraits · 5 points · 30 min · Standard exam level

P05 — The system x' = A(a) x with A = (-4, a; 1, 0): type of the origin for every a, the stable node a = -3 with P, P^{-1}, P^{-1}AP, and the stable focus a = -8 in real form with its sense of rotation

For a real parameter a consider the planar system

$$\dot{\mathbf{x}} = A(a)\,\mathbf{x}, \qquad A(a) = \begin{pmatrix} -4 & a \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

The type of the origin changes with a. Afterwards two values are studied in detail: $a = -3$, with initial condition $\mathbf{x}(0) = (1, 1)^T$, and $a = -8$, with initial condition $\mathbf{x}(0) = (0, 1)^T$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $\mathbf{x}(t) \in \mathbb{R}^2$, $t \in \mathbb{R}$; parameter $a \in \mathbb{R}$.
- Operator: first-order, linear, homogeneous, constant coefficients: solutions exist for all t, are unique, and $\mathbf{x} \equiv \mathbf{0}$ is always an equilibrium. It is the only one when $\det A \neq 0$.
- The type of $\mathbf{0}$ is read from $\tau = \operatorname{tr} A$, $\delta = \det A$ and $\Delta = \tau^2 - 4\delta$ (**extension**: the (τ, δ, Δ) table is derived from $\lambda^2 - \tau\lambda + \delta = 0$; Robinson Ch. 31 lists the cases by type of eigenvalue).

Phase 2. Canonical formulation

$\tau = -4 + 0 = -4$ for every a, $\delta = (-4)(0) - a \cdot 1 = -a$, $\Delta = 16 + 4a$. With an invertible matrix P of (generalised) eigenvectors, $\mathbf{x} = P\mathbf{z}$ turns the system into $\dot{\mathbf{z}} = (P^{-1}AP)\,\mathbf{z}$ with $P^{-1}AP$ diagonal, a Jordan block, or the rotation-dilation block $\begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$.

Part (a) · 1.5 points

Classification for all a

Classify the origin for every real a (table with trace, determinant and discriminant) and describe the borderline cases.

Step 1. Characteristic polynomial

$$\det(A-\lambda I)=\det\begin{pmatrix}-4-\lambda & a\\ 1 & -\lambda\end{pmatrix}=(-4-\lambda)(-\lambda)-a=\lambda^2+4\lambda-a,$$

$$\lambda_{\pm}=\frac{-4\pm\sqrt{16+4a}}{2}=-2\pm\sqrt{4+a}.$$

Step 2. Table

Range of a	τ	$\delta=-a$	$\Delta=16+4a$	**Eigenvalues**	**Origin**
$a>0$	-4	<0	>16	real, opposite signs	saddle (unstable)
$a=0$	-4	0	16	0 and -4	line of equilibria $x_1=0$
$-4<a<0$	-4	>0	$0<\Delta<16$	real, both negative	stable node
$a=-4$	-4	4	0	-2 double	improper stable node
$a<-4$	-4	>4	<0	$-2\pm i\sqrt{-4-a}$	stable focus

For $-4<a<0$: $0<\sqrt{4+a}<2$, so both $-2\pm\sqrt{4+a}$ are negative. For $a>0$: $\sqrt{4+a}>2$, so $\lambda_+>0>\lambda_-$.

Step 3. Borderline values

- $a=0$: $A\mathbf{x}=\mathbf{0}$ means $-4x_1=0$ and $x_1=0$, so every point of the x_2-axis is an equilibrium (eigenvalue 0, eigenvector $(0,1)^T$). For $\lambda=-4$, $(A+4I)\mathbf{v}=\mathbf{0}$ has rows $(0,0)$ and $(1,4)$: $v_1+4v_2=0$, $\mathbf{v}=(-4,1)^T$. Every other orbit is a half-line parallel to $(-4,1)^T$ that ends on the x_2-axis as $t\to\infty$.
- $a=-4$: $A+2I=\begin{pmatrix}-2 & -4\\ 1 & 2\end{pmatrix}\neq 0$; its second row gives $v_1+2v_2=0$, a single eigenvector line through $(-2,1)^T$. The origin is an improper stable node: all orbits approach it tangent to that line.
- $a<-4$: $\rho=-2$, $\omega=\sqrt{-4-a}$. Sense of rotation from the test vector $(1,0)^T$: $A(1,0)^T=(-4,1)^T$, whose second component is positive on the positive x_1-axis, so orbits turn **anticlockwise** (equivalently $a_{12}=a<0$).

Since $\tau = -4 \neq 0$ there is never a centre, and since $\tau < 0$ there is never an unstable node or focus: the only unstable case is the saddle.

KEY RESULTS

- $\lambda^2 + 4\lambda - a = 0, \lambda = -2 \pm \sqrt{4+a}$.
- $a > 0$ saddle; $a = 0$ line of equilibria $x_1 = 0$; $-4 < a < 0$ stable node; $a = -4$ improper stable node, single eigenvector $(-2, 1)^T$; $a < -4$ stable focus, anticlockwise.
- Never a centre and never unstable apart from the saddle.

Part (b) · 1.5 points

The stable node a = -3

For $a = -3$ find eigenvalues and eigenvectors, build P, P^{-1}, check $P^{-1}AP$, write the general solution, solve with $\mathbf{x}(0) = (1, 1)^T$ and describe the phase portrait.

Step 1. Eigenvectors

$A = \begin{pmatrix} -4 & -3 \\ 1 & 0 \end{pmatrix}, \lambda^2 + 4\lambda + 3 = (\lambda + 1)(\lambda + 3)$.

- $\lambda_1 = -1$: $A + I = \begin{pmatrix} -3 & -3 \\ 1 & 1 \end{pmatrix}$. Row 1: $-3v_1 - 3v_2 = 0$; row 2: $v_1 + v_2 = 0$ (the same line). $\mathbf{v}_1 = (1, -1)^T$.
- $\lambda_2 = -3$: $A + 3I = \begin{pmatrix} -1 & -3 \\ 1 & 3 \end{pmatrix}$. Row 1: $-v_1 - 3v_2 = 0$; row 2: $v_1 + 3v_2 = 0$. $\mathbf{v}_2 = (3, -1)^T$.

Step 2. Change of basis

$$P = \begin{pmatrix} 1 & 3 \\ -1 & -1 \end{pmatrix}, \qquad \det P = (1)(-1) - (3)(-1) = 2, \qquad P^{-1} = \frac{1}{2}\begin{pmatrix} -1 & -3 \\ 1 & 1 \end{pmatrix}.$$

Column by column, $AP = [A\mathbf{v}_1 \mid A\mathbf{v}_2]$ with $A\mathbf{v}_1 = (-4 + 3,\ 1)^T = (-1, 1)^T = -\mathbf{v}_1$ and $A\mathbf{v}_2 = (-12 + 3,\ 3)^T = (-9, 3)^T = -3\mathbf{v}_2$:

$$AP = \begin{pmatrix} -1 & -9 \\ 1 & 3 \end{pmatrix}, \qquad P^{-1}AP = \frac{1}{2}\begin{pmatrix} (-1)(-1) + (-3)(1) & (-1)(-9) + (-3)(3) \\ (1)(-1) + (1)(1) & (1)(-9) + (1)(3) \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -2 & 0 \\ 0 & -6 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -3 \end{pmatrix}.$$

Step 3. Solution

In $\mathbf{z} = P^{-1}\mathbf{x}$: $\dot z_1 = -z_1$, $\dot z_2 = -3z_2$, so $z_1 = c_1 e^{-t}$, $z_2 = c_2 e^{-3t}$ (each separable: $\frac{dz_i}{z_i} = \lambda_i\, dt$). Back with $\mathbf{x} = P\mathbf{z}$:

$$\mathbf{x}(t) = c_1 e^{-t}\begin{pmatrix}1\\-1\end{pmatrix} + c_2 e^{-3t}\begin{pmatrix}3\\-1\end{pmatrix}.$$

For $\mathbf{x}(0) = (1,1)^T$: $(c_1, c_2)^T = P^{-1}(1,1)^T = \frac{1}{2}(-1-3,\ 1+1)^T = (-2,1)^T$, so

$$\mathbf{x}(t) = \begin{pmatrix}-2e^{-t} + 3e^{-3t}\\ 2e^{-t} - e^{-3t}\end{pmatrix}, \qquad \mathbf{x}(0) = \begin{pmatrix}1\\1\end{pmatrix}.$$

Step 4. Phase portrait in words

Both eigenvalues are negative: **stable node**. The eigenlines through $(1,-1)^T$ and $(3,-1)^T$ are invariant, with arrows towards the origin. In the $\mathbf{z}$-plane the orbits are $z_2 = K|z_1|^3$ (exponent $\lambda_2/\lambda_1 = 3$), tangent to the z_1-axis at the origin. Back in $\mathbf{x}$: every orbit not on the fast line enters the origin tangent to the **slow** eigenvector $\mathbf{v}_1 = (1,-1)^T$ ($|\lambda_1| = 1$), and far from the origin ($t \to -\infty$) it is parallel to the fast one $\mathbf{v}_2 = (3,-1)^T$. The orbit of the IVP behaves like $-2e^{-t}\mathbf{v}_1 = e^{-t}(-2,2)^T$ for large t: it enters from the second quadrant along $x_2 = -x_1$. Nullclines: $\dot x_1 = 0$ on $x_2 = -\frac{4}{3}x_1$, $\dot x_2 = 0$ on $x_1 = 0$ (there the motion is horizontal, $\dot x_1 = -3x_2$).

KEY RESULTS

- $\lambda_1 = -1, \mathbf{v}_1 = (1,-1)^T$; $\lambda_2 = -3, \mathbf{v}_2 = (3,-1)^T$.
- $P = \begin{pmatrix}1 & 3\\ -1 & -1\end{pmatrix}$, $\det P = 2$, $P^{-1} = \frac{1}{2}\begin{pmatrix}-1 & -3\\ 1 & 1\end{pmatrix}$, $P^{-1}AP = \operatorname{diag}(-1,-3)$.
- $\mathbf{x}(t) = \big(-2e^{-t} + 3e^{-3t},\ 2e^{-t} - e^{-3t}\big)^T$; stable node, tangent to $\mathbf{v}_1$ at the origin.

Part (c) · 2 points

The stable focus a = -8

For $a = -8$ find the complex eigenvalues and an eigenvector $\mathbf{v} = \mathbf{u} + i\mathbf{w}$, build $Q = [\mathbf{u} \mid \mathbf{w}]$, Q^{-1} and check $Q^{-1}AQ$; write the real general solution, solve with $\mathbf{x}(0) = (0,1)^T$ and find the sense of rotation.

Step 1. Eigenvalues and eigenvector

$\lambda^2 + 4\lambda + 8 = 0$: $\lambda = \frac{-4 \pm \sqrt{16-32}}{2} = -2 \pm 2i$. For $\lambda = -2 + 2i$:

$$A-\lambda I=\begin{pmatrix}-2-2i & -8\\ 1 & 2-2i\end{pmatrix}.$$

Row 2: $v_1+(2-2i)v_2=0$; with $v_2=1$, $v_1=-2+2i$. Row 1 checks: $(-2-2i)(-2+2i)-8=(4-4i+4i-4i^2)-8=8-8=0$. So $\mathbf{v}=(-2+2i,1)^T=\mathbf{u}+i\mathbf{w}$ with $\mathbf{u}=(-2,1)^T$, $\mathbf{w}=(2,0)^T$.

Step 2. Action of A on u and w

$A\mathbf{u}=(8-8,\ -2)^T=(0,-2)^T=\rho\mathbf{u}-\omega\mathbf{w}=(4,-2)^T-(4,0)^T$. $A\mathbf{w}=(-8,\ 2)^T=\omega\mathbf{u}+\rho\mathbf{w}=(-4,2)^T+(-4,0)^T$. Both agree with Robinson Eq. 29.3.

Step 3. Change of basis

$$Q=\begin{pmatrix}-2 & 2\\ 1 & 0\end{pmatrix},\qquad \det Q=(-2)(0)-(2)(1)=-2,\qquad Q^{-1}=\frac{1}{-2}\begin{pmatrix}0 & -2\\ -1 & -2\end{pmatrix}=\begin{pmatrix}0 & 1\\ \frac{1}{2} & 1\end{pmatrix}.$$

$AQ=[A\mathbf{u}\mid A\mathbf{w}]=\begin{pmatrix}0 & -8\\ -2 & 2\end{pmatrix}$ and

$$Q^{-1}AQ=\begin{pmatrix}0\cdot 0+1\cdot(-2) & 0\cdot(-8)+1\cdot 2\\ \frac{1}{2}\cdot 0+1\cdot(-2) & \frac{1}{2}\cdot(-8)+1\cdot 2\end{pmatrix}=\begin{pmatrix}-2 & 2\\ -2 & -2\end{pmatrix}=\begin{pmatrix}\rho & \omega\\ -\omega & \rho\end{pmatrix}.$$

Step 4. Real general solution and the IVP

In $\mathbf{z}$: $\dot z_1=-2z_1+2z_2$, $\dot z_2=-2z_1-2z_2$; with $r^2=z_1^2+z_2^2$: $r\dot r=z_1\dot z_1+z_2\dot z_2=z_1(-2z_1+2z_2)+z_2(-2z_1-2z_2)=-2r^2$ and $r^2\dot\theta=z_1\dot z_2-z_2\dot z_1=z_1(-2z_1-2z_2)-z_2(-2z_1+2z_2)=-2r^2$, i.e. $\dot r=-2r$, $\dot\theta=-2$, so $r=r_0e^{-2t}$, $\theta=\theta_0-2t$ (Barrow: $[e^{2s}r(s)]_0^t=0$). Back in $\mathbf{x}$:

$$\mathbf{x}(t)=e^{-2t}\Big[(a_1\cos 2t+b_1\sin 2t)\,\mathbf{u}+(b_1\cos 2t-a_1\sin 2t)\,\mathbf{w}\Big],\qquad (a_1,b_1)^T=Q^{-1}\mathbf{x}(0).$$

For $\mathbf{x}(0)=(0,1)^T$: $(a_1,b_1)^T=(0\cdot 0+1\cdot 1,\ \frac{1}{2}\cdot 0+1\cdot 1)^T=(1,1)^T$. Then

$$\mathbf{x}(t)=e^{-2t}\Big[(\cos 2t+\sin 2t)\begin{pmatrix}-2\\ 1\end{pmatrix}+(\cos 2t-\sin 2t)\begin{pmatrix}2\\ 0\end{pmatrix}\Big]=e^{-2t}\begin{pmatrix}-4\sin 2t\\ \cos 2t+\sin 2t\end{pmatrix}.$$

Check: $\mathbf{x}(0) = (0,1)^T$. By the product rule $\dot{x}_1 = e^{-2t}(8\sin 2t - 8\cos 2t)$, and $-4x_1 - 8x_2 = e^{-2t}\big(16\sin 2t - 8\cos 2t - 8\sin 2t\big) = e^{-2t}(8\sin 2t - 8\cos 2t)$; $\dot{x}_2 = e^{-2t}\big(-2\cos 2t - 2\sin 2t - 2\sin 2t + 2\cos 2t\big) = -4e^{-2t}\sin 2t = x_1$.

Step 5. Rotation and portrait

At $(1,0)^T$, $\dot{\mathbf{x}} = A(1,0)^T = (-4,1)^T$: the velocity on the positive x_1-axis points upward, so orbits turn **anticlockwise** in the $\mathbf{x}$-plane. In the $\mathbf{z}$-plane they turn clockwise ($\dot{\theta} = -2$); the sense flips because $\det Q = -2 < 0$. The origin is a **stable focus**: spirals shrinking by the factor $e^{2\pi\rho/\omega} = e^{-2\pi} \approx 0.0019$ each turn (one turn every π time units). The IVP orbit starts at $(0,1)^T$, moves into the half-plane $x_1 < 0$ and crosses the x_2-axis again at $t = \pi/2$, at $(0, -e^{-\pi})^T$.

KEY RESULTS

- $\lambda = -2 \pm 2i$ ($\rho = -2, \omega = 2$); $\mathbf{v} = (-2+2i,\ 1)^T$, $\mathbf{u} = (-2,1)^T$, $\mathbf{w} = (2,0)^T$.
- $Q = \begin{pmatrix} -2 & 2 \\ 1 & 0 \end{pmatrix}$, $\det Q = -2$, $Q^{-1} = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 1 \end{pmatrix}$, $Q^{-1}AQ = \begin{pmatrix} -2 & 2 \\ -2 & -2 \end{pmatrix}$.
- $\mathbf{x}(t) = e^{-2t}\big(-4\sin 2t,\ \cos 2t + \sin 2t\big)^T$; stable focus, anticlockwise in $\mathbf{x}$ (clockwise in $\mathbf{z}$), radius factor $e^{-2\pi}$ per turn.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** for every $a < 0$ all solutions tend to $\mathbf{0}$ as $t \to \infty$ (asymptotic stability); for $a = 0$ they tend to points of the x_2-axis; for $a > 0$ only the orbits on the stable eigenline reach the origin. As $a \to -4^{\pm}$ the node's two eigenlines merge (from $a > -4$) or the spiral's rotation rate $\omega = \sqrt{-4-a}$ goes to 0 (from $a < -4$): the improper node is the common boundary.
- **Checks:** $P^{-1}AP$ and $Q^{-1}AQ$ were multiplied out; both IVP solutions satisfy the system and the initial data; $\operatorname{tr}(P^{-1}AP) = -4 = \tau$ and $\det(P^{-1}AP) = 3 = \delta$ for $a = -3$, and the same for $a = -8$ (-4 and 8).
- **Common mistakes in the exam:** (1) calling $a = 0$ a node: the origin is not isolated; (2) ordering $Q = [\mathbf{w} \mid \mathbf{u}]$ and then expecting $\begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$ (that order gives the transpose); (3) reading the sense of rotation from the $\mathbf{z}$-plane, which is always clockwise, instead of from a test vector in $\mathbf{x}$; (4) making the node's orbits tangent to the fast eigenvector at the origin.